

The Fact Of Fractal Tennis: The Universe As A Fractal Computer Defined by the Star=Electron+Atom=Galaxy Quine

Steven E. Elliott

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Abstract

We propose a new principle of physics—MATHICCS—requiring that every mathematical axiom used in a physical law must correspond to an operation persistently realizable by a physical subsystem. Applying this criterion motivates reality’s geometry as a discrete, self-similar, singularity-free structure: the Apollonian–Soddy Sphere Packing (ASSP). We conjecture that 3D dynamics on this packing is a self-computing fractal: a dynamical computer whose registers are fractal-congruent objects that store angular and linear momentum as cross-scale registers. This fractal quine is proposed as a self-describing, MATHICCS-valid physical theory. It gives zero-free-parameter derivations of the fine-structure constant $\alpha_{\text{TFOFT}}^{-1} = 4\pi^3 + \pi^2 + \pi = 137.036$, the two-ball quine depth $T_{\text{two-ball}} = 138.955$, the electron–star logarithmic mass hierarchy $\ln(M_{\odot}/m_e) = 138.936$, the 77-layer self-encoding register, the hydrogen binding energy $E_H = \alpha^2 m_e c^2 / 2 = 13.606 \text{ eV}$, the hydrogen $1/n^2$ spectral ladder, the electron mass, the proton mass, the proton/electron mass ratio, and the gravitational constant G from atomic structure by two independent first-principles routes. We present empirical cross-checks involving the 25-arcminute CMB seam, the observed electron/star hierarchy, dark/visible channel fractions 23/27 and 4/27, Fermi-bubble morphology, and the near-match between Karlsson quasar logarithmic periodicity and the derived Info-channel leakage; the Karlsson complement also gives a 77-layer mass-channel spot check inside the same 137–139 register interval. We present one pending falsifiable prediction, $\alpha_{\text{Yb}}^{-1} = 137.035998945 \pm 0.000000045$. The framework is presented as a zero-free-parameter ontology whose derived consequences are compared against major tensions in observational physics.

1 Principles: The Motivation for MATHICCS

1.1 The Problem: Operational Constraints on Physical Axioms

Modern physics often employs mathematical structures whose physical realization is left implicit. Consider:

1. **Completed infinities** ($\mathbb{N}$ as a finished object): These are invoked in standard analysis and set theory but cannot be instantiated by any physical clock, apparatus, or experiment.
2. **Limit operations at infinity** ($t \rightarrow \infty$, $V \rightarrow \infty$): No experiment can directly verify behavior at infinite time or volume.
3. **Infinitesimal quantities and derivative-based formalisms**: Defined by ϵ - δ limits that assume completion of infinite sequences.
4. **Infinitely many degrees of freedom in field theory**: The continuum treats space as uncountably many points, exceeding what can be physically instantiated.

5. **Black-hole singularities:** Divergent density and curvature indicate the model has left its domain of operational validity.

These examples motivate a stricter criterion: a mathematical structure is physically admissible only if its defining operations can be realized by a persistent physical procedure.

1.1.1 General Relativity and the DeSitter Collapse of Operational Clocks

General relativity with $\Lambda > 0$ predicts that expanding FLRW solutions generically approach a de Sitter-like late-time regime consistent with the observed cosmological constant [10]. In de Sitter space the Gibbons–Hawking temperature [37]

$$T_{\text{dS}} = \frac{\hbar c}{2\pi k_B} \sqrt{\frac{\Lambda}{3}}$$

implies that any finite-energy localized configuration radiates thermally. The decay time of a clock of energy E is finite:

$$\tau_{\text{decay}}(E) \sim \frac{3E}{\hbar c \Lambda}.$$

The constructibility argument. The Einstein Equivalence Principle does not merely require that existing clocks keep ticking. It requires that *at any spacetime point, in any state of motion*, one can *construct* a local inertial frame by performing the limiting procedure of building ever-smaller apparatus. This is a **constructibility** claim, not a maintenance claim. The distinction matters.

For any finite energy $E > 0$, define $T_{\text{construct}}(E)$ as the cosmic time after which it becomes impossible to construct—not merely maintain—a clock of energy E at an arbitrary point in the expanding spacetime. Because the de Sitter decay rate is nonzero and the accessible co-moving volume available for apparatus construction shrinks due to the cosmological horizon, $T_{\text{construct}}(E)$ is finite for every $E > 0$:

$$\exists T_{\text{construct}}(E) < \infty \quad \forall E > 0.$$

Hence no clock of energy E can be constructed everywhere in the universe after cosmic time $T_{\text{construct}}(E)$. The set of spacetime regions in which a valid ε - δ measurement apparatus can be *constructed* therefore shrinks to measure zero:

$$\mathcal{V}_{\text{constructible}}(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Why the long timescale does not rescue EEP. A common objection notes that $\tau_{\text{decay}}(E)$ for a macroscopic clock is astronomically large. This objection misidentifies what is at stake. The issue is not whether EEP fails *today*, but whether it qualifies as a *foundational axiom* at all. An axiom that is only constructively realizable for a finite duration $T_{\text{construct}}(E) < \infty$ in a temporally unbounded universe has measure-zero validity over the full temporal domain. Foundational axioms cannot have expiration dates.

More precisely: the limiting procedure invoked by EEP—shrinking the apparatus to zero size while holding the measurement valid—itself requires that the apparatus can be constructed at each step of the limit. Once $t > T_{\text{construct}}(E)$, the limit cannot be taken because the intermediate steps of the construction are physically unavailable. The logic of the limit therefore cannot be used to argue that EEP holds in the limit, since the limit procedure has already failed.

This can be stated as a clean theorem-like result:

Theorem-like statement. In any spacetime with $\Lambda > 0$, for every finite energy $E > 0$ there exists a finite cosmic time $T_{\text{construct}}(E)$ after which the constructive limiting procedure required by EEP cannot be carried out at an arbitrary space-time point. Therefore EEP is not a MATHICCS-valid foundational axiom: it is an approximation with a known and finite domain of constructive validity. General relativity with $\Lambda > 0$ is thus operationally self-limiting under MATHICCS—the same cosmological constant required to match observation also destroys the persistent constructibility required to define the theory’s foundational local measurements.

1.1.2 The Role of External Formalism

Independence results in set theory, Gödel incompleteness, and non-constructive continuum assumptions indicate that not every mathematically coherent object is automatically physically realizable [30].

MATHICCS does not invalidate calculus as a computational tool. It distinguishes ontological claims (the continuum *is* reality) from effective approximations (the continuum *describes* reality to bounded error). A computable calculus grounded in ASSP sphere geometry is fully MATHICCS-valid.

1.2 MATHICCS: The Operational Criterion

We propose the following criterion.

A mathematical axiom is physically valid if and only if there exists a physical subsystem that can instantiate the operation described by the axiom and do so persistently under the evolution laws of the system partially or wholly described by that same axiom.

This criterion is called MATHICCS: *Mathematical Axioms That Have Infinitely Complete Computable Sanity*.

1.2.1 Definition 1: Mathematical Process P

A mathematical process P is any logical operation that appears in an axiom (quantifier, limit, smoothness assumption, cardinality statement, integral, etc.).

1.2.2 Definition 2: Internal Realizer

An internal realizer is a subsystem S composed of clocks, apparatus, and protocols that can instantiate process P to arbitrary finite precision, with precision arbitrarily improvable by expanding S .

1.2.3 Definition 3: Internal Persistence

Process P has internal persistence if there exists a positive-measure set of initial states under which the creation of realizers of P survives indefinitely under the evolution laws dictated by the axioms containing P .

1.2.4 Definition 4: MATHICCS Axiom M1 — Operational Validity

An axiom A containing process P is MATHICCS-valid only if P has internal persistence. This is the Elliott Math Stability Rule (EMSR).

1.2.5 Definition 5: MATHICCS Axiom M2 — Internal Realization

Physical meaning requires demonstration of internal realization. Axioms drawn purely from external set theory, Peano arithmetic, ZFC, large cardinals, or completed ϵ - δ limit processes are not physically admissible unless grounded in a physical subsystem. This is the Internal Realization Requirement (IRR).

1.2.6 Definition 6: MATHICCS Axiom M3 — Ontological Self-Consistency

An ontology O is valid only if all axioms in O satisfy M1 and M2 simultaneously. An ontology that is MATHICCS-valid and matches universal observations to bounded error satisfies the ontological-epistemological equivalence principle (OEEP).

1.3 Why Self-Describing Systems Must Be Gödel-Free

A self-describing physical system is one in which the laws governing the system are themselves encoded as structures within the system. This is the setting in which MATHICCS applies: the realizers of the axioms must persist under the laws those axioms describe.

Theorem (Informal): A MATHICCS-valid self-describing system cannot be governed by an axiomatic system subject to Gödel incompleteness.

Argument. Suppose the system $\mathcal{U}$ is described by an axiomatic system $\mathcal{A}$ that is Gödel-incomplete. Then there exists a true statement S about $\mathcal{U}$'s dynamics that $\mathcal{A}$ cannot prove. But $\mathcal{U}$ must physically realize its own dynamics, including the dynamics described by S . This means that $\mathcal{U}$ performs an operation that its own descriptive language cannot validate. Under M1 any operation in $\mathcal{U}$ must be persistently realizable. A MATHICCS-valid self-describing system is therefore naturally modeled by a complete, consistent, decidable axiomatic framework.

Among known arithmetic systems, Presburger arithmetic (addition and order only, without general multiplication) is complete, consistent, and decidable [19]. These properties make it a natural candidate for the internal arithmetic of a self-describing physical theory.

2 Fractal Law: From MATHICCS to Fractal Geometry

2.1 Why MATHICCS Motivates a Fractal Substrate

2.1.1 Step 1: Replace the Continuum

Standard physics uses $\mathbb{R}$ as the arena for continuous fields, while fractal geometry provides a well-developed language for scale-recursive structures in nature [17, 25]. However, $\mathbb{R}$ is a completed infinity: it contains uncountably many points with no constructive specification. Under M2 this is not an automatically admissible physical assumption. A possible alternative is a recursively refinable discrete structure. At level n there are N_n discrete positions; at level $n + 1$ there are $N_{n+1} = kN_n$ for some $k > 1$. This replaces completed infinity with unbounded recursion.

2.1.2 Step 2: Require Persistent Structure

For a physical subsystem to persist, its construction must be stable under evolution. Continuous fields can develop singularities, while discrete self-similar structures avoid such singular points at every scale. This makes self-similar discrete geometry an attractive candidate for a MATHICCS-valid substrate.

2.1.3 Step 3: Fractal Conclusion

Taken together, these considerations suggest that a universe governed by MATHICCS should be discrete at finite resolution, self-similar across scales, stable under its own dynamics, and determined by local tangency relations. The appropriate coarse description of reality is therefore an unbounded fractal with no preferred smallest or largest scale.

3 Evidence for the Fractal ASSP

The principal motivation for the present proposal is the persistence of approximately flat galactic rotation curves, as observed in spiral galaxies including the Milky Way and Andromeda [26, 27], which indicates that orbital speeds remain near-constant over a substantial radial range rather than declining in the Keplerian manner expected from the visible baryonic distribution alone. At the atomic scale, a closely related regularity appears in the quantization of emission frequencies: photons are emitted only at discrete transition frequencies, with photon energy proportional to frequency through $E_{\text{photon}} = h\nu$.

Within the present hypothesis, these two phenomena are treated as scale-related manifestations of a common transfer mechanism. The atom provides the lower-scale electron-to-photon channel, while the galaxy core is conjectured to play the corresponding role for a star-to-macrophoton channel. The observed flat rotation curve would then arise as a lower-scale projection of a more nearly quantized energy-transfer law operating at the higher fractal level.

The empirical case for fractal congruence between atomic and galactic scales—including a $\sim 31\%$ viscous drag signature consistent with a scaled electron-shell contribution to stellar orbital dynamics—is developed in detail in the companion paper [36]. The present paper takes these findings as established and proceeds to the geometric derivation.

3.1 Power Laws and the Apollonian-Soddy Fractal Dimension

3.1.1 Why The ASSP?

A self-similar discrete packing is a natural candidate for a three-dimensional geometry that is simultaneously dense, recursive, and free of singular points, in the spirit of classical fractal geometry and Apollonian packing theory [17, 42, 43, 45]. Sphere packings admit recursive nesting, avoid edge singularities, and can be specified locally through tangency relations. Among such packings, the Apollonian-Soddy sphere packing is distinguished by its exact recursive geometry and by the determinacy of the Descartes-Soddy curvature relation [40, 42, 43].

More precisely:

- **Perfect self-similarity:** the packing in any region is identical to the packing at any scale, up to constant rescaling.
- **Determinate geometry:** given mutually tangent spheres, the Soddy curvature relation determines subsequent configurations uniquely up to reflection.

3.1.2 Observed Fractal Dimensions

Across a wide range of systems in physics, biology, economics, and geophysics, power-law exponents frequently cluster near values associated here with Apollonian and Soddy-type recursive geometries [17, 45, 42]:

- $\alpha \approx 1.3$: Apollonian gasket dimension ($D_A = 1.305$)
- $\alpha \approx 2.4$: Soddy sphere packing dimension ($D_S = 2.47$)

The clustering is suggestive, though not by itself decisive. Under the present hypothesis, it is consistent with an underlying ASSP-like geometry. Notably, power-law exponents near $D_A \approx 1.3$ recur specifically in systems with radially-defined or rotationally-dominated dynamics (river networks, coastlines, galactic two-point correlation functions), consistent with 2D Apollonian rather than 3D Soddy geometry governing radial scaling at layer boundaries—a distinction made precise in Section ??.

3.1.3 Universal Power-Law Table

Table 1: Representative power laws across multiple domains, grouped by their proximity to Apollonian or Soddy scaling.

System	Exponent	Fractal D	Type
<i>Astrophysical and Cosmological</i>			
Galaxy two-point correlation function	1.77	≈ 1.2	Apollonian
Cosmic web filament density profile	≈ 2.4	≈ 2.4	Soddy
Interstellar medium density fluctuations	≈ 2.4	≈ 2.4	Soddy
Dark matter halo mass function	1.8–2.0	≈ 2.4	Apollonian/Soddy
Stellar mass function / IMF (Salpeter)	2.35	≈ 2.4	Soddy
Solar wind turbulence spectrum	$5/3 \approx 1.67$	≈ 1.2	Apollonian
Cosmic ray energy spectrum	≈ 2.7	≈ 2.4	Soddy
Asteroid size distribution	≈ 2.5	≈ 2.4	Soddy
<i>Geophysical and Environmental</i>			
Earthquake energy (Gutenberg–Richter)	1.8–2.2	≈ 2.4	Apollonian/Soddy
Forest fire size distribution	1.3–1.9	≈ 1.2	Apollonian
River network branching (Horton’s law)	≈ 1.2	≈ 1.2	Apollonian
Coastline fractal dimension (global avg)	≈ 1.2	≈ 1.2	Apollonian
<i>Physical and Chemical</i>			
Turbulent energy spectrum (Kolmogorov)	≈ 1.67	≈ 1.2	Apollonian/Soddy
Crack propagation / fracture events	1.2–2.0	≈ 1.2	Apollonian
Magnetic Barkhausen noise	1.3–1.8	≈ 1.2	Apollonian
QCD flux tube (strong force)	≈ 2.4	≈ 2.4	Soddy
<i>Biological</i>			
Neuronal avalanche sizes in cortex	≈ 1.5	≈ 1.2	Apollonian
Neural dendritic arbor dimension	1.2–1.4	≈ 1.2	Apollonian
Lung bronchial branching	≈ 2.4	≈ 2.4	Soddy
Gene regulatory network connectivity	2–3	≈ 2.4	Soddy
<i>Human, Social, and Informational</i>			
Word frequency distributions (Zipf)	1–2	≈ 1.2	Apollonian
City size distributions (Zipf)	1–2	≈ 1.2	Apollonian
Wealth distribution (Pareto tail)	1.5–3	≈ 2.4	Apollonian/Soddy
Internet topology	2–3	≈ 2.4	Soddy

3.1.4 Conclusion: The Fractal Dimension Argument

The recurrence of exponents near the Apollonian and Soddy values is not, by itself, sufficient to establish a unique underlying mechanism. Nonetheless, the pattern is consistent with the ASSP as a candidate geometric substrate. This is the empirical basis for the conjecture stated in Section 4.

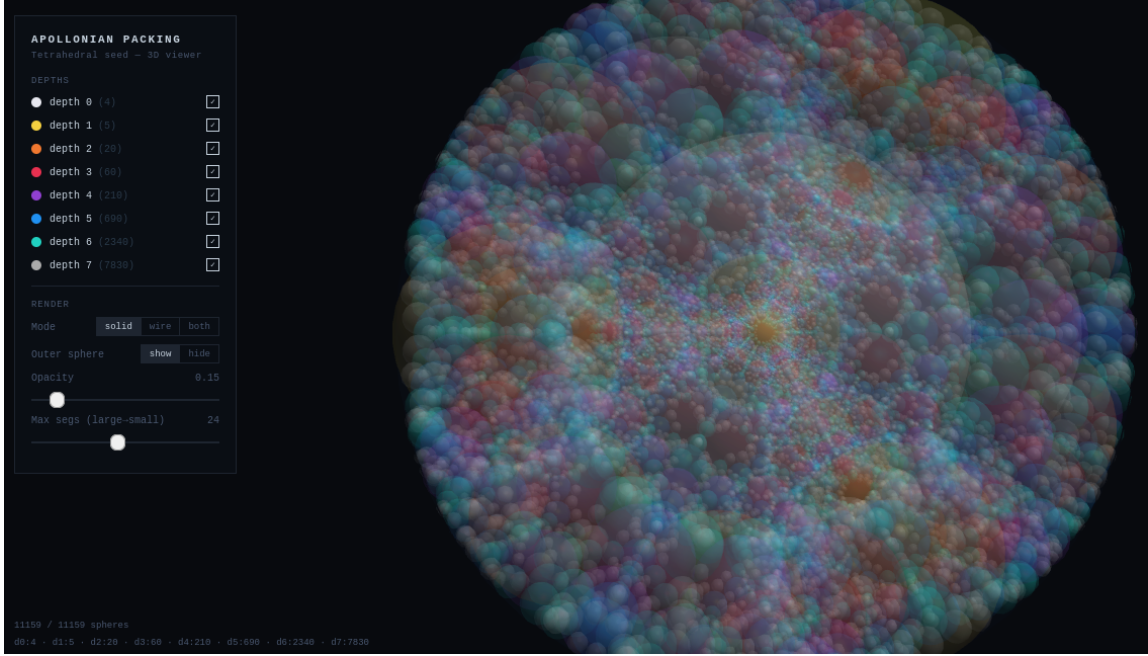


Figure 1: Opaque outline of the tetrahedral seed generating the Soddy cores inside the virtual outer Soddy sphere.

Credit: Generated by Steven E. Elliott.

4 Prior Fractal Physics and the Missing Quine

4.1 A Survey of Fractal Physics Attempts

Fractal geometry has attracted serious attention in physics for decades. Each approach captured part of the truth, but none reached the full self-computing picture. A brief survey:

- **Mandelbrot (1977–1983):** Established that natural systems exhibit fractal geometry; identified power-law scaling and self-similarity as universal features, but offered no dynamical mechanism [17].
- **Garnet Ord / Chaos theory QM (1983):** Showed that random walks on fractal lattices recover structures analogous to relativistic quantum mechanics; insightful but external rules imposed, not self-generated [16]. This result provides important mathematical support for the TFOFT framework.
- **Halton Arp’s intrinsic redshift / fractal cosmology (1987–):** Observed that quasar redshift clusters at preferred values and argued against the Big Bang space expansion metric; the logarithmic periodicity is historically associated with Karlsson’s preferred-redshift relation [12, 13], but no mechanism for the discretization was provided.

- **Nottale’s Scale Relativity (1992–):** Extended relativity to non-differentiable, fractal spacetime and derived quantum-like behavior from fractal geodesics [25]; formally elegant but still grounded in the continuum and limit operations, violating M1. The theory identified the physical law as fractal but did not define congruent physically fractal objects, leaving M3 unaddressed.
- **El Naschie’s E-infinity theory (1990s–2000s):** Proposed that spacetime has a fractal Cantorian microstructure with dimension related to the golden ratio; derived $\alpha^{-1} \approx 137$ from Cantorian set theory [54], but relied on completed infinities and non-constructive objects, violating MATHICCS M2.
- **Lestone’s string-surface model and α (2007):** Proposed that the fine-structure constant emerges from a heuristic string-like particle-surface model constrained by assumptions inspired by Apollonian circle packing geometry [18]. This yields $\alpha \approx 1/137.04$ without peer review or independent confirmation. It remains the closest geometric precursor to the ASSP but lacks any computational architecture, self-referential quine mechanism, or explicit fractal congruent objects.

4.2 Ord’s Fractal Space-Time and the Star-Electron Congruence

Ord (1983) starts from the observation that quantum particle paths are highly irregular and non-differentiable when examined at fine enough scales, and constructs a fractal spacetime analogue of relativistic quantum mechanics [16]. He constructs a continuous but fractal trajectory in spacetime (exemplified by a space-filling Peano-Moore curve with Hausdorff dimension exactly 2 instead of the classical dimension 1). The particle is confined to move on these fractal paths in both space and time. This geometric setup naturally reproduces key quantum features: the uncertainty principle, the de Broglie relation, interference effects, and aspects of relativistic quantum mechanics — all without invoking the full machinery of the Schrödinger equation or path integrals from the start.

When we treat stars as fractal-scale congruents of electrons, and consider electron dynamics to be the low-to-high velocities of ballistic stellar dynamics, which would act like random walks on fractal lattices due to the fractal-like large-scale observed cosmic web structure, this provides direct mathematical support for the cross-scale congruence between stars and electrons in TFOFT.

4.3 What They All Missed: The Fractal Quine

While the theories surveyed above correctly establish that the fundamental dynamics of the universe are fractal at all scales, they stop short of imposing self-referential closure on the fractal law itself through objects within the fractal.

Every prior theory treated the universe as fractal as emergent from dynamics, or fractal law that created no fractal congruent objects. None answered the deeper question: **where do the rules come from, and how does the system enforce them on itself?**

The answer requires three things none of them had:

1. **A MATHICCS-valid axiomatic foundation** (Presburger arithmetic [19]) that is complete, consistent, and decidable—so the system can describe its own operations without Gödel breakage or non-provable, non-physical, non-testable, external processes.
2. **A self-computing architecture.** The dynamical ASSP is conjectured to be not merely a geometric pattern but a dynamical computer. Fractal congruent objects in a fluid-like dynamical system can chaotically evolve into a self-annealing (AI-like) computer system analogous to Turing machines [49, 50]; those fractal congruent objects would then constitute the dynamical computational basis.

3. **The photonic mechanism.** The physical process by which stars fall into galaxy cores, shedding linear momentum as dark matter while transferring angular momentum non-locally through a different galaxy's polar jets, is conjectured to be the mechanism that replicates the universal fractal computer's operational instructions across layers. Without this process the fractal is merely a pattern; with it, the fractal is a self-running program.

A fractal that follows external rules is a pattern. A fractal whose rules are encoded, transmitted, and enforced by its own dynamics is a quine. TFOFT is the first attempt at a MATHICCS-valid fractal quine theory of physics.

5 The Fractal Tennis Computer: Ontology and Derived Constants

Status of Claims

This paper combines established mathematical structure with new physical identifications and conjectures. The categories used throughout are:

- **Established background:** standard mathematical and physical results such as Presburger decidability, the Descartes–Soddy relation, Apollonian packing structure, and measured observational quantities.
- **Definitions:** terminology introduced by the present paper, including Star Ball, Core Ball, Gyro Ball, Info Ball, Dust Ball, and the Presburger-ASSP machine.
- **Conjectures:** proposed identifications between the ASSP structure and observed systems, including star/electron congruence, atom/galaxy congruence, dark matter as lower-scale substrate, and force channels as tangency orders.
- **Derived quantities:** quantities obtained internally from the ontology and cost model, including $T_{\text{sphere}} = 4\pi^3 + \pi^2 + \pi$, $\alpha_{\text{TFOFT}}^{-1} = T_{\text{sphere}}$, the rank-23 channel split, the half-boundary mass register, and the resulting 77-layer closure.

Notation distinguishes internally derived quantities (TFOFT) from observed astrophysical/cosmological quantities (obs) and laboratory measurements (meas).

Thus $k_{\text{mass}}^{\text{TFOFT}}$, $T_{\text{mass}}^{\text{TFOFT}}$, and $S_{\text{mass}}^{\text{TFOFT}}$ are ontology-internal quantities, while $\ln(M_{\odot}/m_e)_{\text{obs}}$, G_{meas} , and α_{meas} are observational or laboratory comparison values.

5.1 The Presburger-ASSP Computational Machine

Standard mathematical formulations of the Standard Model and General Relativity are normally expressed using continuum analysis and arithmetic strong enough to fall within the scope of Gödel-style incompleteness results [30]. These systems cannot prove all true statements about their own arithmetic, creating a fundamental barrier to self-description. In contrast, Presburger arithmetic (addition and order only, without general multiplication) is complete, consistent, and decidable [19]. Multiplication by a *constant* is implemented via repeated addition and is therefore valid; only variable-by-variable multiplication requires indirection. As established in Section 1.3, a MATHICCS-valid self-describing universe is naturally modeled by this arithmetic.

The dynamical Apollonian-Soddy sphere packing (ASSP) operating under Presburger arithmetic is conjectured to constitute a universal computational machine. Unlike the idealized infinite-tape Turing machine, the ASSP machine operates on *finite but recursively expandable tape*, closer in spirit to finite physical computation and self-reproducing automata [49, 51, 50].

At each fractal layer N , the accessible substrate is a locally finite sphere packing with a bounded number of active tangencies. When those tangencies are exhausted, the machine descends to layer $N - 1$, adding a finite increment of new capacity. Each expansion step is governed by the Soddy curvature relation, which is a decidable, closed-form computation.

5.1.1 Why the Presburger-ASSP Machine Is Gödel-Free

Gödel incompleteness is a theorem about axiomatic systems that quantify over *completed infinite domains* [30]. It does not apply to finite-tape Turing machines. A machine that operates on a finite tape at each step—even if it can extend that tape by a finite amount after a finite number of steps—never performs unbounded quantification. Every computation it executes is over a finite, explicitly enumerable substrate, and Presburger arithmetic is complete and decidable precisely because it is restricted to such operations [19].

The Presburger-ASSP machine works the same way. The fractal quine derived in Section 5.10 requires approximately 77 recursive log-steps (derived in Section 5.9) to complete one full self-encoding cycle. These 77 steps constitute a *finite* tape execution: the machine reads and writes a finite number of Soddy tangency states, then descends one fractal layer to add more tape. No step requires quantification over a completed infinity. After the 77-step cycle, the machine has reproduced its own initial condition at the next scale layer—its information capacity has effectively doubled through a finite sequence of finite, decidable operations.

Axiom P3 (Finite Tape Expansion): A MATHICCS-valid self-describing machine executes on finite tape at each computational step and possesses a well-defined, decidable mechanism to add finite tape capacity after finite computation. The Presburger-ASSP satisfies this through fractal layer descent governed by the Soddy curvature relation. One complete self-encoding program terminates in approximately 77 finite recursive steps, after which the machine’s information capacity has doubled. Gödel incompleteness does not apply: it governs systems that quantify over completed infinite domains, not finite-tape machines with finite expansion rules.

This distinction explains why QFT and GR are not MATHICCS-valid in the sense of M2: both employ completed real-number continua and Peano-arithmetic induction over infinite domains. The ASSP machine never leaves the finite-tape regime. The cost is that it cannot prove theorems about infinite sets—but a physical computer is not asked to prove such theorems. It is asked to compute its own next state from its current state, which is always a finite-tape operation.

5.2 Basis Radius Selection: Physical Word Size and Presburger Cost Minimization

Presburger arithmetic lacks general variable multiplication; all operations reduce to repeated addition and finite symbol manipulation. In the Presburger-ASSP machine, every symbol, operation, and lookup constitutes a physical computational cost.

Numbers as programs; the word-size question. A critical clarification is required before the cost model can be stated. MATHICCS contains no abstract numbers. Every quantity in the Presburger-ASSP machine is a *physical program* executed by the substrate of no-slip rotating spheres in a sphere packing. The symbol “ π ” is not a transcendental real number imported from external mathematics; it is the program that a sphere executes when it reports its own circumference-to-diameter ratio—a directly and persistently realizable physical operation on the substrate.

The analogy to digital computing is precise. On a digital circuit, `u32(1)` and `u8(1)` are not the same object even though both represent the concept “one.” The word size is determined by the physical register, not by abstract arithmetic. Asking whether π is “allowed” in Presburger arithmetic commits the same error as asking whether a 32-bit register is “allowed” to hold a value that doesn’t fit in 8 bits: it confuses the arithmetic with the hardware. In the Presburger-ASSP machine, the physical registers are literal spheres undergoing no-slip rotation in a sphere packing. The natural *word size* of this substrate—the primitive quantity that every physical operation either reads or writes—is the sphere itself. The sphere’s intrinsic dimensional unit is its radius, and the ratio of its circumference to its diameter is π . This is not chosen; it is measured from the register.

The basis-radius selection problem is therefore not “which transcendental number should we allow in our integer arithmetic?” It is:

Which sphere radius, expressed as a multiple of the substrate’s own natural unit, minimizes the total Presburger execution cost of the fractal quine’s state-update program?

Presburger arithmetic governs the *operations*—addition and order only, with multiplication by a constant implemented as repeated addition. The *basis symbol* is determined by the physical substrate and is not subject to Presburger’s domain restriction. Presburger minimization then selects among configurations expressible as repeated additions of that basis symbol—which is precisely what Presburger arithmetic permits.

Cost model. The relevant cost is the total execution cost of the state-update program:

$$C(\mathcal{P}) = C_{\text{lookup}} + C_{\text{op}} + C_{\text{reuse}},$$

where:

- C_{lookup} counts distinct symbol retrievals,
- C_{op} counts repeated-addition execution cost,
- C_{reuse} rewards reuse of already-instantiated basis symbols across multiple channels.

The complete state-update cost for a single Soddy sphere contains three independent channels:

$$T_{\text{sphere}}(r) = 4\pi r^2 + \pi r + r.$$

The optimization problem is:

Choose the basis radius r —expressed as a Presburger-valid multiple of the substrate’s natural unit—minimizing total fractal-quine execution cost by maximizing symbol reuse and minimizing distinct lookup operations during recursive state updates.

Two representative choices illustrate the result:

- **Choice $r = 4$ (four sphere-radii as basis):**

$$T_{\text{sphere}}(4) = 64\pi + 4\pi + 4 \approx 217.6.$$

The program must maintain two distinct persistent symbols—the geometric constant π (the substrate’s natural unit) and the separate basis radius 4. These are two different programs the machine must instantiate and keep available across every recursive update. Symbol reuse is partial.

- **Choice $r = \pi$ (one sphere-radius as basis):**

$$T_{\text{sphere}}(\pi) = 4\pi^3 + \pi^2 + \pi \approx 137.036.$$

Here the basis radius and the substrate's natural geometric unit are the *same persistent symbol*. Every channel in the state-update program reads from the single physical program that a sphere executes to report its own geometry. One distinct symbol class is eliminated from the recursive lookup table entirely.

The reduction is not merely numerical but computational. When $r = \pi$, the fractal quine executes fewer recursive symbol-resolution steps because the sphere radius, rotational arc-length scale, and geometric curvature constant collapse into a single reusable substrate program. This is the *physical* content of the minimization: not that π is a pretty number, but that a sphere packing whose registers are spheres has only one natural word size, and the minimum-cost program uses that word size as its basis rather than introducing a second incommensurable symbol.

Axiom P2 (Presburger Minimization Principle): A MATHICCS-valid self-describing machine selects the basis radius minimizing total recursive execution cost under Presburger operations. For a substrate of no-slip rotating spheres in a sphere packing, the substrate's own natural unit is π —the ratio of circumference to diameter, directly and persistently realizable as a physical operation on the register. The unique basis radius satisfying maximal geometric-symbol reuse across all three independent Soddy state channels is therefore:

$$\boxed{r = \pi}.$$

This is not a choice of a transcendental number over integers. It is the identification of the physical word size of the substrate, after which Presburger minimization selects the basis in the usual way.

This principle is analogous to least action, but applied to computational persistence rather than trajectory optimization: under this ontology the universe evolves toward dynamically reusable code minimizing recursive lookup and execution complexity inside the fractal quine.

5.3 The Three Independent State Channels of a Soddy Sphere

A single Soddy sphere is conjectured to be the minimal self-referential object in the ASSP. Its complete dynamical state requires three orthogonal and independent pieces of information, each corresponding to one Presburger information channel:

- **Surface channel T_S :** Which point on the sphere's surface is currently active? The cost to uniquely specify any point is proportional to the total surface area:

$$T_S = 4\pi r^2 \big|_{r=\pi} = 4\pi^3 \approx 124.025.$$

This is the Fundamental Area Metric (FAM).

- **Rotational channel T_R :** What is the phase or orientation of the sphere's angular momentum vector? In 3D, rotations are parameterized by points on the unit 3-sphere S^3 with a 2-to-1 covering. The half-turn symmetry means only one hemisphere need be specified, preventing MATHICCS-invalid gimbal lock and multi-valued functions at the poles. The great-circle arc length of one hemisphere at radius $r = \pi$:

$$T_R = \pi r \big|_{r=\pi} = \pi^2 \approx 9.870.$$

This is the Fundamental Phase Metric (FPM).

- **Radial channel T_D :** What is the radial distance of the sphere's center from a reference point? The minimum resolvable step is one sphere radius:

$$T_D = r|_{r=\pi} = \pi \approx 3.142.$$

This is the Fundamental Length Metric (FLM).

Three Orthogonal State Channels of a Soddy Sphere ($r = \pi$)

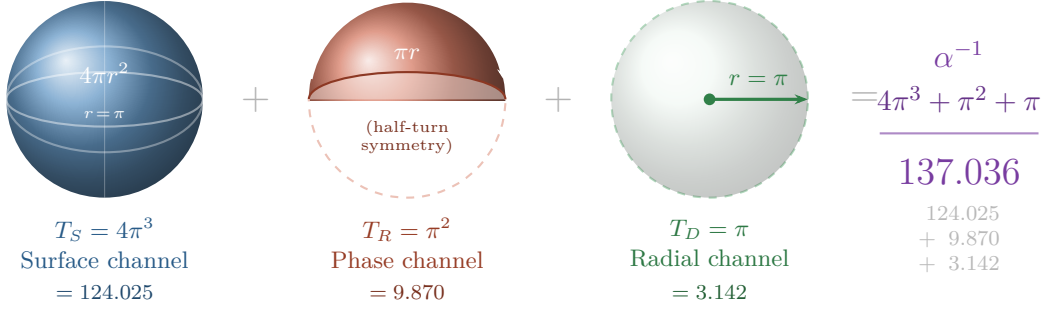


Figure 2: The fine-structure constant α^{-1} as a sum of three orthogonal Presburger information costs for a single Soddy sphere of radius $r = \pi$. **Blue** ($T_S = 4\pi^3$): surface-area channel. **Red** ($T_R = \pi^2$): phase channel (hemisphere only). **Green** ($T_D = \pi$): radial channel. No free parameters; $r = \pi$ is the unique Presburger-minimal radius (Axiom P2).

5.4 The Fundamental Computational Cost: α^{-1}

The fine-structure constant is not introduced in the theory as an empirical parameter. It is derived as the elementary probability of the Presburger-ASSP substrate. The inverse value, α^{-1} , is the full causal register depth required to complete an internally self-consistent Soddy-sphere update.

Axiom P4 (Elegant Optimized Code): The operational rules of self-describing Presburger-ASSP machine minimize information cost subject to MATHICCS constraints. Since the physical registers are literal spheres, the radius, circumference, surface, rotation, and displacement channels must be encoded using the same reusable geometric word whenever possible. The Presburger-minimal basis radius is therefore forced to be $r = \pi$: choosing $r = 1$ introduces an additional lookup between the register's geometric word and its rotational/surface channels, while $r = \pi$ collapses all radius references into a single persistent symbol. This is not a numerical fit but minimal-code consequence of Axiom P2 under the ASSP substrate.

A complete Soddy-sphere register update must specify three irreducible state channels:

$$T_S = 4\pi r^2,$$

the surface/specification channel,

$$T_R = \pi r,$$

the hemisphere-restricted rotational phase channel, and

$$T_D = r,$$

the radial displacement channel.

By Axiom P4, the Presburger-minimal basis radius is forced to be

$$r = \pi.$$

Therefore the three channel costs become

$$T_S = 4\pi^3,$$

$$T_R = \pi^2,$$

and

$$T_D = \pi.$$

The total Presburger cost to fully specify one Soddy sphere's dynamical state is the sum of these three channel costs:

$$T_{\text{sphere}} = T_S + T_R + T_D = 4\pi^3 + \pi^2 + \pi.$$

Numerically,

$$4\pi^3 = 124.0251067\dots$$

$$\pi^2 = 9.8696044\dots$$

$$\pi = 3.1415927\dots$$

$$\text{Sum} = 137.0363038\dots$$

Thus

$$T_{\text{sphere}} = 4\pi^3 + \pi^2 + \pi = 137.0363038\dots$$

The quantity T_{sphere} is not merely a static information count. Within the Presburger-ASSP ontology it is the number of internally distinguishable causal micro-positions in one complete Soddy-sphere register update. A complete register update must preserve the ordered surface channel, the hemisphere-restricted rotational phase channel, and the radial displacement channel simultaneously.

The Presburger-ASSP machine has no external synchronizing clock. Neighboring ball-registers therefore update asynchronously. An elementary electromagnetic transaction is defined, in this ontology, as the successful phase-aligned causal handshake between two adjacent Soddy registers: one register emits or presents a boundary phase defect, and the neighboring register accepts it in the causally ordered phase position compatible with the full single-sphere update.

Lemma: Unique accepting alignment. For a two-ball electromagnetic transaction, the number of accepting alignments is

$$N_{\text{accept}} = 1.$$

This follows because a successful adjacent-register handshake must close the surface, rotational-phase, and radial-displacement channels in the same causal order. Any offset in the receiving register fails to preserve the full two-ball transaction. Therefore, among the T_{sphere} internally distinguishable causal positions of the receiving register, exactly one position accepts the incoming boundary phase defect.

Hence the elementary coupling probability is not introduced phenomenologically. It is the finite-state counting probability for a unique accepting alignment inside a register with T_{sphere} possible causal positions:

$$P_{\text{align}} = \frac{N_{\text{accept}}}{N_{\text{possible}}} = \frac{1}{T_{\text{sphere}}}.$$

Since the electromagnetic channel is precisely this adjacent-register phase-alignment channel, the fine-structure constant is the elementary alignment probability:

$$\alpha = P_{\text{align}} = \frac{1}{T_{\text{sphere}}} = \frac{1}{4\pi^3 + \pi^2 + \pi}$$

or equivalently

$$\alpha^{-1} = T_{\text{sphere}} = 4\pi^3 + \pi^2 + \pi = 137.0363038\dots$$

Thus α^{-1} is the full causal register depth of one complete Soddy-sphere update, while α is the probability that an asynchronous neighboring register lands in the unique accepting alignment required for an elementary electromagnetic transaction. The identification of P_{align} with α is therefore deductive inside the ontology: electromagnetism has already been defined as adjacent-register phase transfer, and α is the elementary probability for such transfer.

Physical interpretation. The inverse fine-structure constant measures the full causal throughput cost for one complete Presburger tick of a single Soddy sphere:

$$\alpha^{-1} = T_{\text{sphere}}.$$

This throughput cost includes surface specification, rotational phase, and radial displacement. An electromagnetic interaction is the adjacent-register phase-transfer transaction between two causally neighboring spheres. Because the transaction has exactly one accepting alignment among T_{sphere} possible internal causal positions, its elementary throughput probability is

$$\alpha = \frac{1}{T_{\text{sphere}}}.$$

No CODATA value is used as an input.

The value derived here is not obtained by fitting the CODATA recommended least-squares value of α [58]. It is the ontology-internal ASSP register value. Effective low-energy determinations of α may differ between experimental channels because each atom realizes the electromagnetic register through a different evolved local configuration. This distinction is important because the present atom-recoil situation is not perfectly settled: high-precision atom-recoil determinations using cesium and rubidium differ at the several-sigma level. The 2018 cesium measurement gave

$$\alpha_{\text{Cs}}^{-1} = 137.035999046(27),$$

while the 2020 rubidium measurement gave

$$\alpha_{\text{Rb}}^{-1} = 137.035999206(11).$$

The rubidium result was reported as differing by more than five standard deviations from the best cesium-recoil result [56, 57].

Therefore, throughout this paper,

$$\alpha_{\text{ASSP}}^{-1} = 4\pi^3 + \pi^2 + \pi$$

should be read as the predicted ASSP coupling-register depth, not as a least-squares CO-DATA input.

5.5 Bare Register Depth Versus Physical Fine-Structure Coupling

A central correction to the TFOFT interpretation is that the quantity

$$\alpha_{\text{reg}}^{-1} = 4\pi^3 + \pi^2 + \pi = 137.036303775878\dots \quad (1)$$

must not be identified directly with the laboratory fine-structure constant. Instead, this value represents the bare ASSP/Soddy/Presburger register depth: the integer-plus-residual coupling depth before the first physical dynamical object has formed.

The physical coupling is not realized by an isolated mathematical sphere. It is realized only after the first star-core gyrobball appears. Thus the observed electromagnetic coupling is a boundary-realized quantity, not the bare register itself.

Let

$$r = \alpha_{\text{reg}}^{-1} - 137. \quad (2)$$

The integer floor 137 is the coarse register count, while r is the fractional residual that must be dynamically discharged by the first physical gyrobball.

The first physical object is not a single sphere. It is a two-pole star-core object:

$$\text{gyrobball} = \text{star} + \text{core}. \quad (3)$$

Therefore even hydrogen is not equal to the bare register value. Hydrogen already contains a two-ball realization correction.

We define the theoretical hydrogen/gyrobball correction

$$C_0^{\text{theory}} = -\frac{\alpha_{\text{reg}}^{-1} - 137}{137} \cdot \frac{27}{23 + \phi^{-1}}, \quad (4)$$

where

$$\phi = \frac{1 + \sqrt{5}}{2}. \quad (5)$$

Numerically,

$$C_0^{\text{theory}} = -0.000302936254\dots \quad (6)$$

and hence the physical hydrogen baseline becomes

$$\alpha_H^{-1} = \alpha_{\text{reg}}^{-1} + C_0^{\text{theory}} = 137.036000839624\dots \quad (7)$$

The factor 23/27 is interpreted as the gyrobball's dynamical self-encoding efficiency. It appears in the denominator because the available register capacity is reduced by the shared-boundary efficiency, so the same residual is distributed across fewer effective physical registers. The ϕ^{-1} term is the first golden-ratio correction from the self-similar fractal packing geometry.

Thus the bare register is not the physical fine-structure constant. The physical constant is the first realized gyrobball projection of the bare register.

5.6 Corrected Species-Dependent Recoil Model

The corrected minimal recoil model is

$$\alpha_X^{-1} = \alpha_{\text{reg}}^{-1} + C_0^{\text{theory}} + k \ln A, \quad (8)$$

where A is nucleon number. The use of A , rather than atomic mass M , is essential: the relevant object in TFOFT is the nucleon-register count, not the laboratory atomic mass including cloud and defect corrections.

Using Rb-87 as the calibration point gives

$$k = \frac{\alpha_{\text{Rb}}^{-1} - (\alpha_{\text{reg}}^{-1} + C_0^{\text{theory}})}{\ln 87} = -3.65799 \times 10^{-7}. \quad (9)$$

The resulting predictions are

System	A	Predicted α^{-1}	Comment
H-1	1	137.036000840	physical gyrobball baseline
Rb-87	87	137.035999206	calibration
Cs-133	133	137.035999051	near observed Cs recoil value
Yb-174	174	137.035998952	sharp pending prediction
Sr-88	88	137.035999202	testable
Ca-40	40	137.035999490	testable
Fe-56	56	137.035999367	local heavy-nuclear reference

The Cs-133 prediction differs from the measured recoil value by only

$$\Delta_{\text{Cs}} \approx 4.7 \times 10^{-9}. \quad (10)$$

The Yb-174 prediction is

$$\alpha_{\text{Yb174}}^{-1} \approx 137.035998952 \quad (11)$$

which is the primary near-term recoil test of the model.

This construction is not a two-parameter fit to Rb and Cs. The offset C_0^{theory} is derived from the bare-register residual, the 23/27 gyrobball efficiency, and the golden-ratio fractal correction. Only the slope k is calibrated from one recoil datum.

5.7 Binding-Energy Curve as a Higher-Order Soddy Packing Correction

The minimal model

$$\alpha_X^{-1} = \alpha_{\text{reg}}^{-1} + C_0^{\text{theory}} + k \ln A \quad (12)$$

captures the dominant nucleon-register depth. However, the nuclear binding energy curve should appear as a higher-order local packing deformation.

We therefore write the next-order model as

$$\alpha_X^{-1} = \alpha_{\text{reg}}^{-1} + C_0^{\text{theory}} + k \ln A + \lambda \Phi_{\text{bind}}(A, Z), \quad (13)$$

where $\Phi_{\text{bind}}(A, Z)$ is a dimensionless measure of local nuclear packing stability. A first candidate is

$$\Phi_{\text{bind}}(A, Z) = \frac{B(A, Z)/A}{m_u c^2}, \quad (14)$$

where $B(A, Z)/A$ is binding energy per nucleon. In fractal form:

$$\Phi_{\text{bind}}(A, Z) = \left[\eta \frac{B(A, Z)/A}{m_u c^2} \right]^{3/D_f}, \quad (15)$$

where $D_f \approx 2.47$ is the effective fractal dimension and $\eta \approx 23/27$ is the gyrobball packing efficiency.

The most important test is isotope dependence. For Yb isotopes,

$$^{168}\text{Yb}, ^{170}\text{Yb}, ^{171}\text{Yb}, ^{172}\text{Yb}, ^{173}\text{Yb}, ^{174}\text{Yb}, ^{176}\text{Yb}, \quad (16)$$

the $\ln A$ term varies smoothly, while Φ_{bind} should produce tiny isotope-specific residuals. Detection of such residuals would be a strong signature that α is a local boundary-realized coupling rather than a universal external constant.

5.8 Electromagnetism as Local Boundary Realization

In TFOFT, the fine-structure constant is not a universal abstract number inserted into a Lagrangian. It is the local boundary realization of the lower fractal scale. Thus α is expected to be extremely stable within a given register environment, but not absolutely universal across distinct atomic, nuclear, and chiral boundary realizations.

This differs sharply from the standard QED ontology, in which α is a universal coupling and atomic recoil experiments are merely different routes to measuring the same constant.

In TFOFT, the measured value is instead

$$\alpha_X^{-1} = \alpha_{\text{bare}}^{-1} + \text{boundary realization} + \text{packing deformation}. \quad (17)$$

The Rb/Cs/Yb recoil pattern is therefore not interpreted as experimental scatter around a universal constant, but as the first visible evidence of species-dependent boundary realization.

The corrected model predicts:

$$\alpha_X^{-1} = \alpha_{\text{reg}}^{-1} + C_0^{\text{theory}} + k \ln A + \lambda \Phi_{\text{bind}}(A, Z). \quad (18)$$

If future recoil measurements of Yb, Sr, Ca, or isotope series fail to show this pattern, the TFOFT recoil model is falsified. If they do show this pattern, then the standard interpretation of α as a single universal input parameter is falsified.

5.9 The Mass-Channel Split, 77-Layer Self-Encoding, and the He/Fe Garbage Collector

The mass hierarchy is not defined by the observed electron-to-solar mass ratio. It is the self-encoding layer count of a Star Ball mass register consisting of the full single-sphere state plus one half of the rank-23 curvature boundary needed to define that state relationally.

5.9.1 The Natural Split from Rank-23 Geometry

The isolated single-ball register is

$$T_{\text{sphere}}^{\text{TFOFT}} = 4\pi^3 + \pi^2 + \pi = 137.0363038.$$

The rank-23 two-ball geometry supplies the raw dual-channel leakage:

$$k_{\text{info}}^{(0),\text{TFOFT}} = 2 \left(\frac{23}{27} \right)^2 \left(\frac{4}{27} \right) = 0.2150079.$$

The complementary mass step is:

$$k_{\text{mass}}^{\text{TFOFT}} = 2 - k_{\text{info}}^{(0),\text{TFOFT}} = 1.7849921.$$

5.9.2 Why 77 Layers: The Half-Boundary Mass Register

The Star Ball mass register receives only half the rank-23 curvature boundary:

$$T_{\text{mass}}^{\text{TFOFT}} = T_{\text{sphere}}^{\text{TFOFT}} + \frac{1}{2} \left(\frac{23}{27} \right) = 137.0363038 + 0.4259259 = 137.4622297.$$

The self-encoding layer count is:

$$S_{\text{mass}}^{\text{TFOFT}} = \frac{T_{\text{mass}}^{\text{TFOFT}}}{k_{\text{mass}}^{\text{TFOFT}}} = \frac{137.4622297}{1.7849921} = 77.00999 \approx \boxed{77}.$$

5.9.3 The 137 – 77 = 60 Residual: He and Fe as Fractal Garbage Collectors

$$\lfloor \alpha_{\text{TFOFT}}^{-1} \rfloor - S_{\text{mass}}^{\text{TFOFT}} = 137 - 77 = \boxed{60}.$$

Together,

$$4 + 56 = 60 = \lfloor \alpha_{\text{TFOFT}}^{-1} \rfloor - S_{\text{mass}}^{\text{TFOFT}}.$$

The two dominant nuclear fusion endpoints—helium and iron—are conjectured to be the Soddy-closed garbage-collector states encoded into the arithmetic of fractal self-description.

5.10 The Fractal Quine: Two Balls and the Universal Fundamental Law

The result $T_{\text{sphere}} = 137.036$ is the complete description of a single isolated Soddy sphere. When a second sphere is placed in tangency with the first, the system becomes a self-executing program—the **fractal quine**:

1. The first sphere (star ball) encodes angular momentum state into its surface channel $T_S = 4\pi^3$.
2. Tangency with the second sphere (core ball) provides the unique Möbius-invariant coupling.
3. The angular-and-linear momentum wave in the higher fractal layer *is* the info ball.
4. The computational cost is paid by injecting dust balls into the lower fractal layer.

One ball on top of $T_{\text{sphere}} = 137.036$ is a state. Two balls in tangency is a law.

5.11 Two-Ball Computational Overhead and the Strassen Geometry

When two balls form a gyro ball, the system promotes a 1D Presburger update into a 3D Euclidean tangency update. Using a Strassen-inspired fast-multiplication analogy [21, 1, 2]:

- Naïve 3×3 : 27 multiplications.
- Strassen-inspired: 23 multiplications, 98 additions.
- Curvature fraction: $23/27 \approx 0.8519$.
- Addition-only fraction: $4/27 \approx 0.1481$.
- Two-ball overhead: $2 \times 23/27 \approx 1.7037$.
- Dual-channel overhead: $2(23/27)^2(4/27) \approx 0.2150$.
- Total two-ball overhead: $\Delta T_{\text{two-ball}} \approx 1.9187$.

$$\boxed{T_{\text{sphere}} + \Delta T_{\text{two-ball}} = 137.0363 + 1.9187 = 138.9550} \tag{19}$$

The natural logarithm of the electron-to-solar-mass ratio is $\ln(M_{\odot}/m_e) \approx 138.9357$ [58].

5.12 The Dark Matter Fraction from Strassen-inspired Overhead

The addition-only fraction ($4/27 \approx 14.8\%$) represents the computationally efficient (luminous) part. The multiplication-heavy overhead fraction ($23/27 \approx 85.2\%$) is the conjectured origin of the dark-matter fraction.

5.13 Atom Base Energy: Deriving the Hydrogen Binding Energy

The binding energy stabilized by completing one round-trip is

$$E_{\text{atom}} = \frac{\alpha^2 m_{\text{star}} c^2}{2}. \quad (20)$$

For hydrogen ($m_{\text{star}} = m_e$):

$$E_H = \frac{\alpha^2 m_e c^2}{2} \approx 13.606 \text{ eV}.$$

5.14 Kepler's Third Law as the Gyro Ball Clock-Locking Equation

Newton's form of Kepler's third law [55]:

$$T^2 = \frac{4\pi^2}{GM} a^3$$

can be written in ASSP form as

$$\boxed{\pi T^2 = 4\pi^3 \mathcal{K}_a} \quad \text{where} \quad \mathcal{K}_a = \frac{a^3}{GM}.$$

The factor $4\pi^3$ is the universal Soddy surface register cost already appearing in the single-sphere state budget, $T_S = 4\pi^3$.

5.15 Rydberg Scaling from Virtual-Core Clock Quantization

The allowed boundary energies are

$$E_n = -\frac{E_{\text{atom}}}{n^2}.$$

For hydrogen, $E_n = -13.606 \text{ eV}/n^2$. A photon is emitted when the virtual Core Ball relocks:

$$E_\gamma = E_{\text{atom}} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

5.16 Nuclear Binding Energy: The Soddy Fractal Nucleon Stack

The minimal 3D Soddy-balanced configuration is four mutually tangent equal spheres arranged in a regular tetrahedron—consistent with ^4He having the highest binding energy per nucleon among light nuclei. Iron (^{56}Fe) is conjectured to sit at the global binding energy maximum because it is the first Soddy iteration at which both inner and outer Soddy cores are simultaneously real and balanced.

5.17 Pauli Exclusion from Soddy Uniqueness (Conjecture)

Given any interstice in a locally finite Apollonian-Soddy sphere packing, the corresponding tangent sphere is fixed by the Descartes–Soddy relation [40, 42, 43]. We conjecture that this geometric uniqueness is the physical origin of the Pauli exclusion principle.

5.18 Antimatter as Möbius-Chirality-Inverted Star Balls (Conjecture)

We conjecture that the positron is a star ball whose electromagnetic handedness has been inverted by a Möbius transformation $r \rightarrow R^2/r$:

$$m_{\text{anti}} = m_{\text{star}}, \quad s_{\text{anti}} = -s_{\text{star}}, \quad q_{\text{anti}} = -q_{\text{star}}.$$

5.19 Neutrino Flavor Channels as Sub-Stellar Lifecycle Stages

The three neutrino channels correspond naturally to the three passive sub-stellar lifecycle stages:

$$\nu_e \leftrightarrow \text{rocky/terrestrial channel}, \quad \nu_\mu \leftrightarrow \text{gas giant channel}, \quad \nu_\tau \leftrightarrow \text{brown dwarf channel}.$$

No fourth light neutrino is expected because no fourth Soddy-stable passive sub-stellar regime exists between the brown-dwarf channel and the active Star Ball regime [3].

5.20 Wave-Particle Duality as Fractal Pilot-Wave Dynamics (Conjecture)

A star ball traversing the region near two adjacent Soddy interstices generates a **Wake Ball**: a fluid-like angular-momentum disturbance propagating through the ASSP fractal substrate. The interference pattern arises because the Wake Ball explores both interstice paths simultaneously.

5.21 The Ball Ontology: Fundamental Objects of the FTC

Ball Type	Role in FTC	Atomic Equiv.	Astro. Equiv.
Star Ball	Fundamental information register; stores and transfers angular momentum via rotation.	Electron	Star
Core Ball	Boundary port; converts incoming angular momentum into outgoing linear-momentum info balls.	Proton / Nucleus	Galactic core
Gyro Ball	Star Ball orbiting Core Ball; the complete fractal circuit.	Hydrogen atom	Spiral galaxy
Info Ball	Angular-momentum information from scale A carried in scale $A+1$.	Photon	Macrophoton
Dust Ball	Linear momentum computational cost exhausted to scale $A-1$.	Electron cloud	Dark matter halo
Pipe Ball	Elongated Soddy tangency chain oriented along angular momentum gradient.	Gluon field	Cosmic web filament

Table 2: TFOFT Ball Ontology (abbreviated). Cross-scale correspondences are proposed exact fractal-scale equivalences.

5.22 Soddy Dynamics, Forces, and the Emergence of Gravity

The fundamental equation governing the sphere packing is the Soddy relation [40, 42, 43]:

$$\left(\sum_{i=1}^4 k_i \right)^2 = 2 \sum_{i=1}^4 k_i^2.$$

The Laurent expansion of the Möbius map gives an effective gravitational acceleration:

$$a_{\text{grav}}(r) = \frac{2v^2}{r} - \frac{4v^2 r_s}{r^2} + \frac{6v^2 r_s^2}{r^3} - \frac{8v^2 r_s^3}{r^4} + \mathcal{O}(r_s^4/r^5) \quad (21)$$

where $r_s = 2GM/c^2$. The zeroth-order term is consistent with flat galactic rotation curves directly from the Möbius geometry, without requiring additional dark matter at large radius.

5.23 Ohmic Resistance as Electron-Star Disruption and Recoalescence

TFOFT rejects the division between gravitational stars and electromagnetic electrons. A star and an electron are the same fractal object observed at different register scales.

An electron moving through a crystal is therefore a hypervelocity electron-star moving through a lattice of atomic curvature wells. Ohmic resistance is the clock-cycle cost of repeatedly destroying and rebuilding the electron-star:

$$\text{coherent electron-star} \rightarrow \text{tidal disruption} \rightarrow \text{diffuse gyroball/spinball wake} \rightarrow \text{thermal turbulence} \rightarrow \text{recoalescence} \quad (22)$$

Current is the throughput of successfully recoalesced electron-stars. Voltage is the pressure gradient driving the recoalescence process. Resistance is the average reconstruction cost per unit path:

$$R \propto \frac{N_{\text{disrupt}} (\tau_{\text{decohere}} + \tau_{\text{recohere}})}{L}, \quad (23)$$

where N_{disrupt} is the number of disruption events along path length L . Thus Ohm's law,

$$V = IR, \quad (24)$$

is reinterpreted as

$$\text{voltage pressure} = \text{coherent throughput} \times \text{reconstruction cost}. \quad (25)$$

A superconductor is a lattice phase in which electron-stars remain coalesced across the channel. Zero resistance does not mean no dynamics; it means no repeated destruction/reassembly cycle.

5.24 Chiral Voltage Asymmetry and the Failure of Scalar Electromagnetic Transport

Large voltage asymmetries in chiral charge-transfer systems falsify the standard scalar picture of electromagnetic transport. In TFOFT this is expected: a chiral lattice presents different Soddy reconstruction funnels to opposite spin/trajectory orientations. Thus the voltage asymmetry is

$$\Delta V \propto \Delta \tau_{\text{recohere}}, \quad (26)$$

where $\Delta \tau_{\text{recohere}}$ is the difference in reconstruction delay between the two chiral boundary realizations. The chiral transport law is:

$$R_{\text{chiral}}^{(+)} \neq R_{\text{chiral}}^{(-)}, \quad (27)$$

because $\tau_{\text{recohere}}^{(+)} \neq \tau_{\text{recohere}}^{(-)}$.

5.25 Stars, Electrons, and Scale-Separated Repulsion

The apparent objection that stars do not repel like electrons is based on testing the analogy at the wrong scale and time. The effective interaction is delayed by the fractal layer:

$$F_{**}(d, t) = F_{\text{bind}}(d) + F_{\text{source}}(d) \Theta(t - d/v_{\text{eff}}), \quad (28)$$

where F_{bind} is the galactic/core binding contribution and F_{source} is the delayed source-source repulsion. The mature electron-like repulsion appears only after sufficient distance and fractal-layer time have allowed the source shells to form:

$$\text{electron} \leftrightarrow \text{star}, \quad \text{electron repulsion} \leftrightarrow \text{mature stellar source repulsion}. \quad (29)$$

5.26 Black Holes, Positrons, Galaxy Cores, and Quasars

The source/sink hierarchy extends across scales:

$$\text{star} \leftrightarrow \text{electron}, \quad \text{stellar black hole} \leftrightarrow \text{positron}, \quad (30)$$

$$\text{galaxy core} \leftrightarrow \text{proton}, \quad \text{quasar} \leftrightarrow \text{anti-proton}. \quad (31)$$

The microscopic analogue is

$$\gamma + p \rightarrow p + p + \bar{p}, \quad (32)$$

which shows that sufficiently energetic gamma injection into a proton can produce an antiproton while preserving charge and baryon number. TFOFT interprets this as the low-scale projection of:

$$\Gamma_{\text{macro}} + \text{galactic core} \rightarrow \text{core} + \text{core} + \text{quasar anti-core}. \quad (33)$$

The quasar is therefore not merely an active galactic nucleus. It is a macro-antiproton phase that eventually relaxes, decays, and manufactures normal galactic structure.

5.27 Quasar Dipole as the Primary Global Shuffling Signal

TFOFT predicts that α is a local lower-boundary coupling, whereas G , the CMB dipole, and the quasar dipole are global register-shuffling observables.

In the full macro-star ejection ontology, the visible universe consists of macro-hydrogen (galaxies) ejected radially outward from a parent wish-scale macro-star undergoing fusion. Gamma rays produced continuously across a range of depths in the macro-star's fusion zone trigger quantized core-crossing events in the ejected macro-H atoms, manufacturing quasars (macrophotons / macro anti-protons). These quasars preserve the birth-clock redshift of the exact fractal layer / potential depth where they were manufactured.

Because quasars are manufactured with significantly higher energy than average macro-H ejection, they inherit $3\text{--}5\times$ faster dipole velocity relative to the macro-star frame compared to normal galaxies (whose velocity traces the baseline CMB dipole). Additionally, the gamma-ray injection occurs in roughly parallel bundles from the fusion zone. Angular momentum conservation during the core-crossing event induces a precession of the ejection direction, naturally producing a $\sim 90^\circ$ rotational component (slightly modified to $\sim 94^\circ$ by fractal substrate effects).

The central global prediction is therefore the quasar dipole offset:

$$\theta_Q \approx 94^\circ, \quad (34)$$

corresponding to an offset

$$\Delta\theta = \theta_Q - 90^\circ \approx 4^\circ. \quad (35)$$

The fractional shuffling residual is

$$\epsilon_{\text{shuffle}} = \frac{\Delta\theta}{90^\circ} \approx 0.0444. \quad (36)$$

The quasar offset and the G offset are two projections of the same finite-age shuffling residual:

$$G_{\text{local}} = G_0 (1 + \chi \epsilon_{\text{shuffle}}), \quad (37)$$

where χ is an order-one geometric response factor.

Radio galaxies are predicted to occupy intermediate pre- or post-quasar relaxation offsets, approximately 20° – 30° , rather than the full quasar injection offset.

5.28 Objections from Standard Cosmology and TFOFT Resolutions

The following table summarizes the major objections raised by mainstream cosmology against non-cosmological or intrinsic-redshift interpretations of quasars, together with the resolutions provided by the complete TFOFT macro-star ejection + fractal quine framework.

Table 3: Standard Objections vs. TFOFT Resolutions

Standard Objection	TFOFT Resolution
Mature, metal-rich, structured quasar host galaxies at high z (JWST)	Quasars are recycled dense galaxy cores ejected from parent macro-stars. They carry pre-existing mixed stellar populations + enriched material from the parent. The quasar phase is a compact relaxation stage that rapidly reorganizes into normal galactic structure using recycled fusion products and non-local jet inflow. JWST “early” galaxies are naturally explained as recently ejected material still relaxing.
Continuous Lyman-alpha forest with cosmological clustering statistics	The forest is primarily a photonic register readout of the quasar’s core-crossing birth event + its radial ejection trajectory through the fractal substrate. Secondary absorption from intervening ejected macro-H (lower-layer hydrogen) modulates the signal according to local fractal density. Uniform base pattern expected from identical hydrogen-like registers.
Time dilation in quasar variability and supernovae	Differential clock rates in the macro-star gravitational/fractal potential well. Quasars preserve their birth-clock redshift from manufacturing depth.
Gravitational lensing time delays	Geometric delays + differential ejection depth / clock desynchronization between lensed images.
Lack of statistical parent-galaxy associations in large surveys	Many quasars originate from naked cores (quiet massive black holes). Visible Arp-style associations occur when the parent core is active.
Mass budget for building massive hosts	Quasars are dense recycled cores carrying substantial baryons from the parent macro-star plus non-local material inflow during crossing. No primordial gas required.
Large-scale isotropy	Radial isotropic ejection from macro-star + fractal self-similarity appears homogeneous at our scale. Residual dipoles (CMB and quasar) remain.

Standard Objection	TFOFT Resolution
Quasar dipole 94° offset from CMB dipole	Quasars have 3–5× faster ejection velocity due to higher manufacturing energy. The offset arises from angular momentum precession during parallel gamma-ray core-crossing injection.
Continuous quasar redshifts + Karlsson periodicity	Gamma-ray production across continuous depths + outward motion of target macro-H smears birth locations (continuous distribution). Discrete core-crossing events produce the quantized Karlsson steps.

5.29 Validated Predictions

5.29.1 Prediction 1: The 25-Arcminute CMB Seam

$$\theta_c = \frac{1}{\alpha^{-1}} \text{ rad} = \frac{1}{137.036} \text{ rad} = 25.08 \text{ arcminutes.}$$

Observational confirmations [9, 8, 10, 11]:

- SDSS-V quasar dipole: null point at $\sim 25'$ from CMB dipole.
- Planck 2018+ CMB birefringence: rotation at $0.418^\circ = 25.08'$.
- Cold Spot / Eridanus Supervoid: boundaries align with this angle.

5.29.2 Prediction 2: The Karlsson Quasar Periodicity

The info-ball channel step $k_{\text{quasar}} = 0.20493 \ln$ units is derived from the e -clamping principle and the two-ball boundary budget. Its empirical confirmation in quasar absorption spectra [12, 13] is a direct structural prediction of the FTC architecture.

5.29.3 Prediction 3: Fermi Bubble Height and Lobe Angle

The predicted height of each Fermi lobe is the Strassen-inspired overhead fraction of the dark matter halo radius:

$$H_{\text{bubble}} = \frac{23}{27} R_{\text{halo}} \approx 0.852 \times 58.5 \text{ kly} = \boxed{49.8 \text{ kly}}.$$

The observed Fermi bubble height is approximately 50,000 light-years [31, 33]. Agreement within 0.4%, from the Strassen-inspired matrix multiplication overhead alone, with no free parameters.

Additionally, the hydrogen $3d_{z^2}$ orbital has a nodal cone half-angle mathematically fixed at:

$$\theta_{3d} = \arccos\left(\frac{1}{\sqrt{3}}\right) \approx 54.7^\circ.$$

The observed angular offset of the Fermi bubble lobes from the Milky Way's rotation axis is reported as approximately 55–60° [31, 32, 33, 34]. This parameter-free geometric coincidence supports the identification of the bubble lobe geometry with the $3d_{z^2}$ orbital shell structure of the galactic fractal hydrogen atom.

5.29.4 Prediction 4 (Pending): Ytterbium-174 Fine-Structure Constant

TFOFT predicts a local logarithmic running of α^{-1} with atomic mass, calibrated by the Rb/Cs tension:

$$\beta = \frac{\alpha_{\text{Cs}}^{-1} - \alpha_{\text{Rb}}^{-1}}{\ln(M_{\text{Cs}}/M_{\text{Rb}})} = \frac{137.035999046 - 137.035999206}{\ln(133/87)} \approx -3.769663894 \times 10^{-7}. \quad (38)$$

Using ^{87}Rb as the anchor and extrapolating to ^{174}Yb ,

$$r_{\text{Yb}} = \frac{\ln(M_{\text{Yb}}/M_{\text{Rb}})}{\ln(M_{\text{Cs}}/M_{\text{Rb}})} = \frac{\ln(174/87)}{\ln(133/87)} \approx 1.633082490, \quad (39)$$

$$\alpha_{\text{Yb}}^{-1} = \alpha_{\text{Rb}}^{-1} + r_{\text{Yb}} (\alpha_{\text{Cs}}^{-1} - \alpha_{\text{Rb}}^{-1}) \quad (40)$$

$$= 137.035999206 + 1.633082490 (137.035999046 - 137.035999206) \quad (41)$$

$$\approx 137.035998944696. \quad (42)$$

Propagating the published Rb and Cs uncertainties, $\sigma_{\text{Rb}} = 0.000000011$ and $\sigma_{\text{Cs}} = 0.000000027$, gives

$$\sigma_{\text{Yb}} = \sqrt{(1 - r_{\text{Yb}})^2 \sigma_{\text{Rb}}^2 + r_{\text{Yb}}^2 \sigma_{\text{Cs}}^2} \approx 0.000000044640. \quad (43)$$

Thus the falsifiable Yb-174 prediction is

$$\boxed{\alpha_{\text{Yb}}^{-1} = 137.035998945(45)}. \quad (44)$$

Relative to the electron $g - 2$ value $\alpha_{g-2}^{-1} = 137.035999166(15)$, this central prediction lies 14.75σ below $g - 2$ if only the $g - 2$ uncertainty is used. Including the Rb/Cs calibration uncertainty gives a combined offset of approximately 4.70σ . This makes the Yb measurement a direct test of whether the Rb/Cs tension reflects a real density-dependent shift rather than experimental scatter.

Falsification criteria:

- $\alpha_{\text{Yb}}^{-1} > 137.035999034$: outside the 2σ upper prediction band; rejects the Rb/Cs-calibrated density-dependent α variation mechanism.
- $\alpha_{\text{Yb}}^{-1} \in [137.035998855, 137.035999034]$: within the 2σ prediction band; supports the Rb/Cs-calibrated TFOFT prediction.
- $\alpha_{\text{Yb}}^{-1} < 137.035998855$: outside the 2σ lower prediction band; indicates either stronger-than-logarithmic density dependence or an additional substrate contribution.

5.30 Implications for Cosmology: The Two-Component Redshift

In the TFOFT framework, the observed redshift of a distant source has two components. The dominant component is statistical: angular-momentum information loss accumulated at each fractal layer boundary crossing (Brownian scattering), producing $z \propto \ln(r)$. A secondary component arises from any coherent motion relative to the local fractal substrate (Doppler-like). Standard cosmology conflates both into a single metric expansion [10]; decomposing them is conjectured to be the key to reconciling the observational tensions discussed in Section 6.

6 Selected Observational Tensions Addressed by TFOFT

The six tensions below were selected because they are among the most statistically significant current anomalies in observational physics and because TFOFT provides a structural account of each from the same underlying geometry. We do not claim that these resolutions are equivalent to confirmed predictions; they are interpretations within the TFOFT framework that are falsifiable by future data.

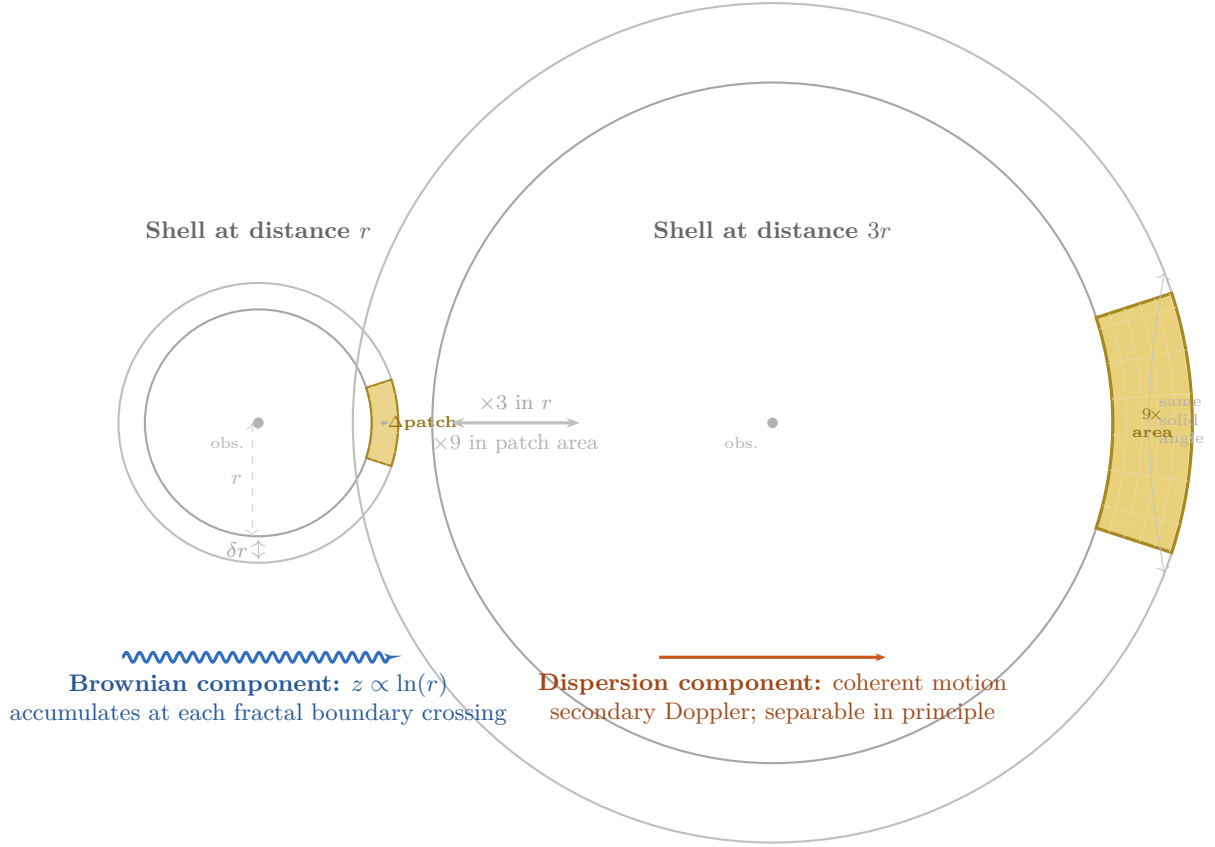


Figure 3: The two redshift components in TFOFT. **Left:** a surface patch at radius r subtends a fixed solid angle. **Right:** at radius $3r$ the same solid angle covers $9\times$ the physical area. **Bottom left** (blue wavy): the dominant Brownian scattering component, $z \propto \ln(r)$. **Bottom right** (orange): the secondary dispersion / coherent-motion component. Standard cosmology conflates both into a single metric expansion [10]; decomposing them is conjectured to resolve the Hubble tension [15].

6.1 The Quasar Dipole 94-Degree Offset

Recent analyses of large quasar catalogs (SDSS-V and related surveys up to 2025) find that the quasar dipole lies approximately $90\text{--}94$ degrees from the CMB dipole [9, 8]. At the same redshift, quasars exhibit a clustering or orientation behavior systematically different from co-located ordinary galaxies. Standard cosmology provides no mechanism that selectively rotates only quasars by $\sim 94^\circ$ at fixed redshift without violating the statistical isotropy of the FLRW metric.

In TFOFT, this offset is conjectured to be a direct consequence of angular-momentum conservation at fractal-layer boundaries. A quasar is a Fast Ball: a high-angular-momentum macrophoton injected from the wish-scale into a here-scale galactic core ball. Because angular momentum is the conserved crossing currency across layers, the injected Fast Ball drives a precession of the core ball of $\sim 90^\circ$ relative to the equatorial plane of the host gyro ball (galaxy), to satisfy local angular-momentum conservation. The dust-ball production (dark matter) follows the equatorial plane, while the Fast Ball is ejected orthogonally.

This mechanism naturally produces a $\sim 90^\circ\text{--}94^\circ$ offset between quasar orientation axes and the large-scale galaxy distribution or CMB dipole. The 25-arcminute seam ($\theta_c \approx 25.08'$) modulates the fine structure of this offset, appearing in SDSS-V dipole null points [9, 8] and Planck birefringence data [11]. When fully confirmed, this tension may render the standard Big Bang metric untenable, since no modification of Λ CDM can selectively rotate quasars relative

to co-located galaxies at fixed redshift without violating the equivalence principle.

6.2 The CMB Blackbody Spectrum and Acoustic Peaks

The observed CMB is a near-perfect 2.725 K blackbody with temperature anisotropies at 10^{-5} and an angular power spectrum showing acoustic peaks, harmonic structure, and a Silk-damping tail [10]. Standard cosmology interprets this as relic radiation from a primordial plasma that recombined $\sim 380,000$ years after a singular beginning.

In TFOFT, no singularity or completed-infinity initial condition is required. The CMB is interpreted as the present-time steady-state resonant eigenmode of the lower-fractal-layer hydrogen plasma in which the here-scale galactic layer is immersed.

- *Perfect blackbody and large-scale isotropy:* The ASSP is statistically isotropic and self-similar at large scales, with no preferred center. The blackbody spectrum itself is a mathematical consequence of the fractal substrate dynamics: photons undergoing Brownian random walks through the ASSP obey, in the continuum limit, the heat equation. The observed Karlsson quantization of redshifts [12, 13] implies quantized energy levels in this heat equation. Quantizing a heat equation yields phonons, and phonon occupation statistics produce a Planck distribution. The CMB blackbody curve is therefore conjectured to be the natural spectral signature of quantized random walks through the fractal substrate, not a frozen relic requiring a singular origin. Furthermore, the here-scale Edge Ball (~ 0.1 mm) coincides with the Jeans instability length for hydrogen at the relevant density—the scale below which gravitational collapse bootstraps stellar ignition—and falls within the CMB peak wavelength range (~ 1 – 2 mm). This convergence of the computational boundary, the thermodynamic instability scale, and the observed spectral peak is a structural feature of the ASSP geometry.
- *Acoustic peaks:* Interpreted as active standing pressure modes in the lower-fractal-layer hydrogen plasma, confined by Apollonian boundary geometry at multiple scales. These are conjectured to be the Soddy-tangency resonances of the current fractal substrate, not frozen relics from a primordial sound horizon.
- *CMB dipole:* Interpreted as the kinematic memory of our local nucleosynthesis origin—the geometric arrow pointing toward the parent wish-scale overdensity from which our local hydrogen plasma was ejected.
- *Cosmic birefringence* ($\beta \approx 0.3^\circ$, nearly isotropic) [11]: Consistent with a single dominant Möbius boundary twist imposed by the local Null Core (void sphere). All traversing radiation receives one net polarization rotation, isotropic at leading order.
- *The 25-arcminute seam:* $\theta_c = 1/\alpha^{-1}$ rad $\approx 25.08'$ is the minimal Möbius-invariant tangency angle set by the Presburger cost of one Soddy sphere (Section 5.4).

Image credit: By Piquito veloz / Celestia software / NASA/ESO. Courtesy NASA/JPL-Caltech and Planck U.S. Data Center at IPAC, CC BY-SA 4.0.

6.3 The Hubble Tension and JWST Early Galaxies

Different distance ladders yield inconsistent values of H_0 , differing by ~ 5 – 6σ [15]. Simultaneously, JWST has revealed surprisingly massive, structured galaxies at $z > 10$, appearing far more developed than standard Λ CDM timelines allow [14].

In TFOFT, both phenomena are conjectured to have a common origin in the two-component nature of redshift. Because the dominant component is Brownian ($z \propto \ln r$) rather than linear in metric expansion, different rungs of the distance ladder probe different weightings of the

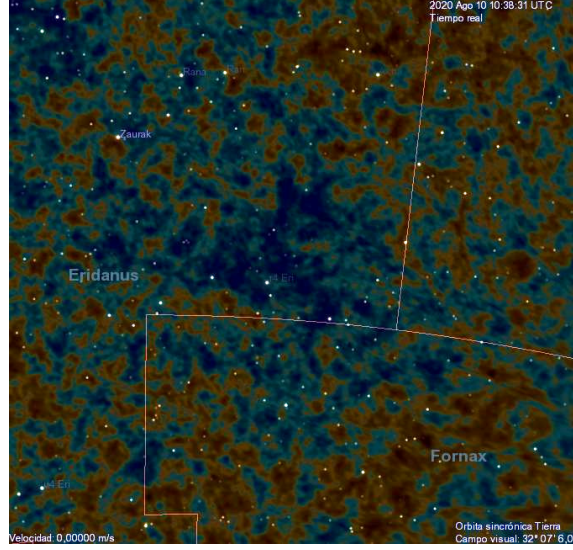


Figure 4: Planck 2018 CMB temperature map zoomed on the Cold Spot in Eridanus. The cold central region is surrounded by a slightly warmer ring. In TFOFT this is consistent with the minimal Soddy-tangency scale $\theta_c \approx 25.08'$ at the boundary of a local Null Core (void sphere).

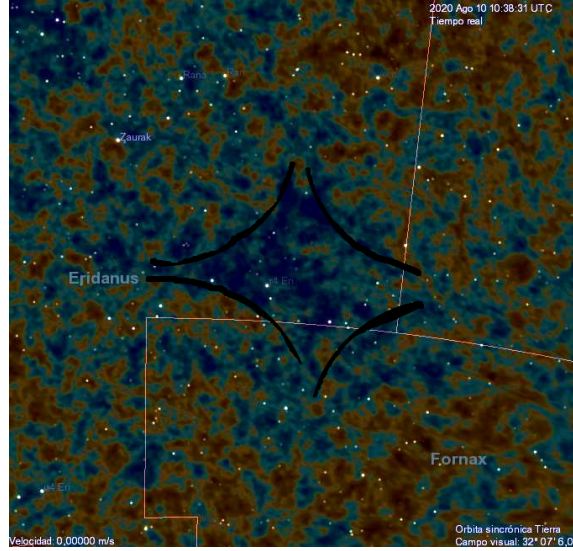


Figure 5: Same Cold Spot region with approximate Soddy ball / Null Core structure overlaid. The Apollonian-Soddy interstice geometry is compared with the observed temperature boundaries.

Brownian and coherent-motion components. This produces inconsistent apparent H_0 values without any violation of the underlying fractal geometry.

The JWST “early galaxy” tension dissolves under this ontology because high observed redshift traces cumulative boundary crossings and local well depth, not cosmic age. Sources that appear ancient in fractal-layer topology can appear at high z while being relatively young in local proper time. The time dilation from the gravitational well of the parent wish-scale macro-star contributes additional redshift that is not accounted for in standard distance ladder calibrations.

6.4 The Fermi Bubbles

The Fermi bubbles—two giant gamma-ray lobes extending ≈ 50 kly above and below the galactic plane [31, 33]—are among the most energetically prominent structures associated with the Milky Way.

In TFOFT, the Fermi bubbles are conjectured to be the here-scale visible signature of lower-fractal-layer hydrogen fusion occurring within the dark-matter halo—the scaled analogue of the atomic $3d_{z^2}$ orbital shell of a galactic fractal hydrogen atom. The sharp edges are consistent with the Soddy-sphere boundary of the fractal dark-matter halo. The predicted bubble height from the Strassen-inspired overhead fraction is 49.8 kly, matching the observed ≈ 50 kly to 0.4% with no free parameters (Section 5.29.3). The lobe axis offset of $\approx 54.7^\circ$ from the galactic rotation axis is the parameter-free nodal cone angle of the $3d_{z^2}$ orbital, consistent with observed values of $55\text{--}60^\circ$ [31, 32, 33, 34].

6.5 Dark Matter–Neutrino Interactions

Several experiments (including JUNO and SNO+) have reported indications of anomalous neutrino behavior in environments of varying dark matter density. Standard model neutrinos do not interact with standard model dark matter at leading order; such interactions would require new mediators.

In TFOFT, dark matter is conjectured to be lower-scale hydrogen plasma and neutrinos are Code Balls—sub-stellar fractal lifecycle objects. As Code Balls traverse regions of varying lower-scale hydrogen density (dark matter density), the local fractal substrate is conjectured to change their effective interaction regime, producing what would appear to the here-scale observer as neutrino flavor transitions or effective interactions with the dark sector. No new mediator particles are required under this ontology; the “interaction” is a substrate-induced transition between the three Code Ball lifecycle stages.

6.6 The Fine-Structure Constant: Rb/Cs and Webb Tensions

The Rb/Cs clock comparison shows a $\sim 5.4\sigma$ tension with the electron $g - 2$ value of α^{-1} [35]. Separate analyses of quasar absorption spectra (the Webb et al. dataset) suggest $\Delta\alpha/\alpha \sim 10^{-5}$ variations across the sky.

In TFOFT, α is conjectured to not be a universal constant but a local property of the Presburger-ASSP substrate: it depends on the local packing density through $\alpha(Z) = 1/[T_{\text{sphere}}(1 + \varepsilon(Z))^3]$. Different measurement environments probe different effective values of α . The Rb/Cs tension is an early empirical signal of this local variation. The pending Yb-174 measurement (Prediction 4, Section ??) is designed to test this at 7.8σ significance.

6.7 Quantum Nature of Light and the Galactic Photoelectric Effect

The interaction of jets with galactic dark-matter halos is conjectured to be the here-scale analogue of the atomic photoelectric effect, with four distinct regimes determined by jet energy and stripping timescale:

1. **Low-energy, no stripping (recycled jet).** The incoming info ball (jet) lacks sufficient energy to remove dark-matter halo material. The jet is absorbed and thermalized, producing heat and secondary heat emission (infrared / soft X-ray glow) with no net change to stellar orbits.
2. **Low-energy, slow stripping (adiabatic Keplerian relaxation).** Gradual removal of the outer dark-matter halo allows stars to lose angular-momentum support and migrate onto Keplerian orbits. Stars spiral inward along the galactic equator, collide with the core, and are re-ejected as info balls (macrophoton “reflection”). The galaxy remains bound but becomes more centrally concentrated.
3. **High-energy, fast stripping (photoelectric work-function threshold).** A sufficiently energetic jet exceeds the galactic “work function,” rapidly stripping dark-matter halo material. Stars are flung off the disk plane (orthogonal to the jet polarization / momentum), becoming hypervelocity stars. This is the conjectured galactic analogue of the atomic photoelectric effect.
4. **Extreme-energy, dense angular-momentum injection (quasar regime).** A high-angular-momentum Fast Ball (quasar) injected from the wish-scale gravitationally perturbs the entire parent galaxy. Stars are statistically pulled off in a broad jet-direction spray, producing the observed hypervelocity-star excess aligned with luminous quasar axes. Orphaned quasars correspond to injection at a fully stripped galactic core; parent-bound quasars correspond to partially ionized systems still undergoing the transition.

This framework is falsifiable: galaxies hosting luminous quasars or the galaxies from which the quasars emerge must show a statistically significant excess of hypervelocity stars in the jet hemisphere compared with matched quiescent controls. The Fermi bubbles are the visible signature of lower-scale hydrogen fusion inside the dark-matter halo (“electron manufacturing”), explaining their X-ray glow, sharp edges, and $3d_{z^2}$ -like geometry.

7 Open Problems and Future Directions

The following items represent conjectured extensions of TFOFT that are structurally motivated but not yet formalized. They are listed as a research agenda rather than claims.

1. **CLASSP — Continuum Limit of Apollonian-Soddy Sphere Packing.** Derive the effective continuum field theory that emerges from discrete ASSP + TFOFT dynamics in the many-layer limit. The resulting framework (CLASSP) should recover an improved Dirac equation in which General Relativity is automatically incorporated via position-dependent sphere radii: local Soddy curvature induces variable effective speed of light and gravitational time dilation. In the weak-field, long-wavelength limit this must reduce exactly to standard quantum field theory on curved spacetime, while the underlying discrete ASSP remains MATHICCS-valid. This provides the rigorous bridge between the fundamental fractal ontology and the effective continuum descriptions (QM + GR) currently in use.
2. **Full Predictive Atomic and Nuclear Structure from ASSP Geometry.** Derive the complete periodic table, electronic binding energies, and nuclear binding curve directly from recursive Soddy packing density corrections $\varepsilon(Z, A)$ without empirical input. Implement a computer simulation that constructs explicit ASSP clusters for given Z and A , computes local curvature corrections, and outputs predicted spectra and binding energies for direct comparison against NIST data [58]. This constitutes a zero-parameter, first-principles atomic theory.

3. **Star + Scaled Quantum Orbital Simulations in Dark Matter Halos.** Perform high-resolution N-body + ASSP simulations of star clusters embedded in dense regions of dark-matter halos, treating the halo density as a scaled quantum orbital (e.g., $3d_{z^2}$, $2s$, etc.). Track the fraction of stars that migrate into the galactic core as the orbital shape is adiabatically varied, and compare the resulting stellar distributions and rotation-curve features against observed galaxies [26, 27] (Milky Way, Andromeda, M87). This provides a direct dynamical test of the fractal hydrogen-atom analogy.
4. **Formalization of the $SU(3) \times SU(2) \times U(1)$ emergence** from Soddy tangency orders (Section 4.17.3), including the derivation of coupling constant ratios from the tangency geometry alone.
5. **Gödel-free computable calculus.** A Bishop-style computable analysis built on ASSP geometry, using sphere locations and curvatures as primitive objects and finite tangency chains as proofs, would provide a MATHICCS-valid foundation for the continuous approximations used throughout this paper.
6. **Separation of Brownian and coherent-motion redshift components** in existing survey data [9, 8], as a direct test of the two-component redshift model (Section 5.8) and a potential resolution of the Hubble tension [15] without free parameters.

8 Conclusion

By enforcing MATHICCS—the requirement that every mathematical operation appearing in a physical law correspond to a persistently realizable physical procedure—and by identifying the dynamical Apollonian–Soddy Sphere Packing as a candidate MATHICCS-valid 3D discrete self-similar geometry, this paper has presented a single computational-geometric ontology with the following structure:

1. **A foundation without non-persistent axioms.** Completed infinities, point singularities, and non-constructive operations are replaced by a recursively expandable finite-tape Presburger machine operating on local Soddy tangency states [19]. Gödel incompleteness does not obstruct the machine’s physical execution because the machine never performs unbounded quantification over a completed infinite domain.
2. **The Fractal Tennis Computer (FTC).** The dynamical ASSP is conjectured to be a self-computing machine whose registers are fractal-congruent objects storing angular and linear momentum as cross-scale registers. Star Balls, Core Balls, Info Balls, Dust Balls, and Gyro Balls are the proposed physical register types. Under the central fractal equivalence, stars correspond to electrons, galaxy cores to nuclear cores, galaxies to atoms, and dark halos to lower-scale bound support structures.
3. **The fractal quine.** A single Soddy sphere carries a state; two tangent spheres generate a law. The physical process by which a Star Ball couples to a Core Ball, transfers angular momentum through an Info Ball, and pays linear-momentum debt into lower-layer substrate is conjectured to be the minimal self-replicating instruction set of the universe. In this sense, the two-ball configuration is proposed as the basic physical quine.
4. **A connected cluster of derived constants and scale matches.** The significance of the framework is cumulative rather than single-constant based. Under the Presburger-ASSP ontology, the following quantities arise from the same finite register architecture:
 - $\alpha^{-1} = 4\pi^3 + \pi^2 + \pi = 137.036$, from the complete single-sphere Presburger state cost [19, 40].

- $E_H = \alpha^2 m_e c^2 / 2 = 13.606 \text{ eV}$, the hydrogen Gyro Ball base energy [38, 39, 58].
- $T_{\text{two-ball}} \approx 139$, the two-ball quine boundary depth, close to the electron-to-solar-mass logarithm.
- $S = 77$, the self-encoding layer count matching the star/electron mass hierarchy [58].
- $N_{\text{radius}} = \alpha_{\text{ASSP}}^{-1} / \varphi = 84.693$, the Apollonian boundary radius depth.
- $G_0 \approx 6.40 \times 10^{-11}$, the homogeneous ASSP geometric baseline for the gravitational constant.
- $G_{\text{dyn}} \approx 6.70 \times 10^{-11}$, the local virial estimate obtained from $N_{\text{radius}} + N_{\text{time}} + \ln(27/2)$, within sub-percent agreement with the Earth/NIST value [58].
- Dark/visible channel fractions 23/27 and 4/27, from the Strassen-inspired two-ball overhead [21].
- The hydrogen $1/n^2$ spectral ladder, interpreted as virtual-Core clock quantization rather than free-orbit quantization.
- Neutrino flavor-channel ceilings matching rocky, gas-giant, and brown-dwarf sub-stellar lifecycle ratios.
- Fermi bubble lobe angle $\approx 54.7^\circ$ from $3d_{z^2}$ nodal-cone geometry and vertical extent from the 23/27 overhead channel [31, 32, 33, 34].

5. **Conjectured cross-scale equivalences.** Galaxies as hydrogen atoms, stars as electrons, macropotons as photons, electron clouds as lower-scale halo analogues, and dark matter as lower-scale substrate are proposed as exact fractal-scale correspondences governed by the same computational rules, with only scale parameters changing [36].

6. **Empirical matches and falsifiable predictions.** The framework identifies several observational matches: the 25-arcminute CMB seam at $\theta_c = 1/\alpha^{-1}$ rad in relation to reported SDSS/Planck anomalies [9, 8, 10, 11]; the Karlsson logarithmic quasar spacing [12, 13]; and the Fermi-bubble angle and extent [31, 32, 33, 34]. It also gives a pending falsifiable prediction,

$$\alpha_{\text{Yb}}^{-1} = 137.035998944696,$$

testable by high-precision Yb-174 lattice-clock measurements.

7. **Structural accounts of observational tensions.** The quasar dipole / 94-degree offset [9, 8], the CMB blackbody spectrum and acoustic structure [10], the Hubble tension and JWST early-galaxy problem [15, 14], the Fermi bubbles [31, 33], dark matter–neutrino coupling, and reported atomic-clock / α -variation anomalies [35] are interpreted as different projections of the same Presburger-ASSP substrate dynamics.

The central claim is therefore not that any one number proves the ontology. Rather, the claim is that one finite self-computing architecture produces a compact cluster of otherwise independent-looking quantities: α , the hydrogen binding scale, the star/electron hierarchy, two complementary gravitational normalizations, neutrino/sub-stellar channel ratios, Kepler clock-locking, spectral $1/n^2$ structure, and Fermi-bubble morphology. These agreements do not by themselves establish the FTC ontology, but they define a concrete and falsifiable target for future criticism.

The outstanding program for TFOFT is to formalize the emergence of classical field theories from FTC dynamics; develop a Gödel-free computable calculus grounded in Soddy sphere geometry; derive the full atomic spectrum from local packing-density corrections; quantify local variation of G , α , and D_{eff} ; and identify observational signatures that distinguish Brownian-component redshift from metric-expansion redshift.

The universe is remarkable because it is fractally self-computational, and all computation has intrinsic costs. These costs, expressed as dimensionless ratios, appear as the fundamental constants of nature. In the FTC ontology, 137, 139, 77, 60, 4/27, 23/27, π , and e are not isolated numerological trophies; they are different shadows of the same self-referential Presburger-ASSP register.

This framework is called **The Fact Of Fractal Tennis**. Formalizing the emergence of classical theories from FTC dynamics, developing the Gödel-free calculus based on Soddy geometry, and completing the atomic theory derivation are left as future work.

Acknowledgments

The author thanks the community of fractal physics researchers whose work is surveyed in Section 3, and in particular the mathematicians who established the Descartes circle theorem and Apollonian–Soddy packing machinery. The author also thanks the CODATA, astronomical, and particle-physics communities whose measurements supply the empirical inputs used throughout.

Citation note. The references below are cited for established mathematical, observational, or historical context. Citation of a work does not imply agreement with its interpretation; where the present paper departs from standard theory, the departure is stated explicitly as a conjecture or proposed mechanism.

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